

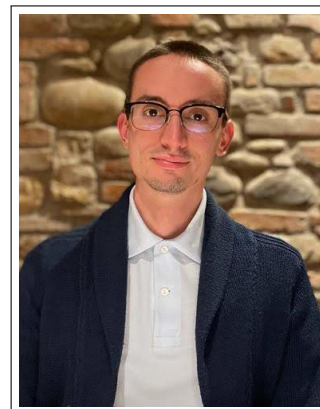
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Born on October 9, 1993 in Piacenza, Italy.
Nationality: Italian.

Current position

Assistant professor (biträdande universitetslektor), SU.

Areas of specialization

Numerical mathematics for partial differential equations, approximation theory, model order reduction, frequency-domain applications, scattering problems.

Appointments held

- 2014–2017 [*Special courses teacher*, Piacenza (IT).
- 2016 [*Developer intern*, Iren SpA, Piacenza (IT).
- 2017–2021 [*Doctoral assistant*, EPFL, Lausanne (CH).
- 2022 [*Post-doctoral researcher*, EPFL, Lausanne (CH).
- 2022–2024 [*University assistant and post-doctoral researcher*, University of Vienna, Vienna (AT).
- 2024–2025 [*Post-doctoral researcher*, KTH, Stockholm (SE).
- 2025–now [*Assistant professor*, SU, Stockholm (SE).

Education

- 2012–2015 [BSc in Applied Mathematics (*cum laude*), Politecnico di Milano, Milan (IT).
Thesis: “A mathematical justification of the momentum operator in quantum mechanics”.
Advisor: Prof. M. Verri.
- 2015–2017 [MSc in Computational Science and Engineering, EPFL, Lausanne (CH).
Project: “Implementation of smooth contact mechanics with the mortar method”.
Advisor: Prof. G. Ancaux.
Project: “Finite elements-based Padé approximants for Helmholtz frequency response problems”.
Advisor: Prof. F. Nobile.

Thesis: “Randomized low-rank approximation of matrices and tensors”.

Advisor: Prof. D. Kressner.

2017–2021 [PHD in Mathematics, EPFL, Lausanne (CH).

Thesis: “Model order reduction based on functional rational approximants for parametric PDEs with meromorphic structure”.

Advisor: Prof. F. Nobile.

Grants, honors, and awards

2011 [3rd place at the “Hong Kong International Science Fair”.

2013 [4th place in the “Championnat International des Jeux Mathématiques et Logiques”.

2014 [5th place in the “Championnat International des Jeux Mathématiques et Logiques”.

2017 [Douchet prize for best GPA, MATH-EPFL.

2020 [Prize for exceptional teaching service, Section of Mathematics, EPFL.

2021 [Junior Research Fellowship at ESI Vienna.

2025 [NAISS small compute grant for study of U-Kolmogorov Arnold networks.

Publications

Journal articles

2019 [F. Bonizzoni and DP, “Distributed sampling for rational approximation of the acoustic scattering of an airfoil”, PAMM 19.

2020 [F. Bonizzoni, F. Nobile, I. Perugia, and DP, “Fast Least-Squares Padé approximation of problems with normal operators and meromorphic structure”, Math. Comput. 89.

F. Bonizzoni, F. Nobile, I. Perugia, and DP, “Least-Squares Padé approximation of parametric and stochastic Helmholtz maps”, Adv. Comput. Math. 46.

[DP, “Interpolatory minimal rational model order reduction of parametric problems lacking uniform inf-sup stability”, SIAM J. Numer. Anal. 58.

2021 [F. Bonizzoni and DP, “Shape optimization for a noise reduction problem by non-intrusive parametric reduced modeling”, Proc. WCCM-ECCOMAS2020.

DP and F. Nobile, “Frequency-domain non-intrusive greedy Model Order Reduction based on minimal rational approximation”, Sci. Comput. Electr. Eng. 36.

[F. Nobile and DP, “Non-intrusive double-greedy parametric model reduction by interpolation of frequency-domain rational surrogates”, ESAIM:M2AN 55.

2022 [DP and F. Nobile, “A technique for non-intrusive greedy piecewise-rational model reduction of frequency response problems over wide frequency bands”, J. Math. Ind. 12.

2023 [F. Bonizzoni, DP, and M. Ruggeri, “Rational-approximation-based model order reduction of Helmholtz frequency response problems with adaptive finite element snapshots”, Math. Eng. 5.

DP, “Adaptive approximation of nonlinear eigenproblems by minimal rational interpolation”, PAMM 22.

[DP, “Toward a certified greedy Loewner framework with minimal sampling”, Adv. Comput. Math. 49.

2024 [P. Huwiler, DP, and J. Schiffmann, “Plug-and-play adaptive surrogate modeling of parametric nonlinear dynamics in frequency domain”, Int. J. Num. Meth. Eng. 125.

[DP and A. Borghi, “Match-based solution of general parametric eigenvalue problems”, J. Comp. Phys. 519.

- 2025 [DP, M. Nonino, and I. Perugia, “Geometry-based approximation of waves in complex domains”, SIAM J. Appl. Math. 85.
 J. Heiland, I.V. Gosea, U.Römer, DP, H. Sreekumar, and S. Langer, “Adaptive rational interpolation and higher-order SVD for low-rank tensor approximation in structural dynamics simulations”, 2025 ECC Proc.
 [DP, R. Hiptmair, and I. Perugia, “Surrogate modeling of resonant behavior in scattering problems through adaptive rational approximation and sketching”, J. Sci. Comp. 104.
 2026 [DP, I. V. Gosea, and J. Heiland, “Barycentric rational approximation for learning the index of a dynamical system from limited data”, J. Sci. Comp. 107.

Pending articles

- 2025 [DP, A. Borghi, L. Pieronek, and A. Kleefeld, “Bifurcations in interior transmission eigenvalues: theory and computation”, under review.

Talks and attendance at events

Presentations at conferences

- 2019 [DP, F. Nobile, F. Bonizzoni, and I. Perugia, “A technique for rational model order reduction of parametric problems lacking uniform inf-sup stability”, GAMM Annual Meeting 2019, Vienna (AT).
 DP, F. Nobile, F. Bonizzoni, and I. Perugia, “A technique for rational model order reduction of parametric problems lacking uniform inf-sup stability”, ICIAM 2019, Valencia (ES).
 [DP and F. Nobile, “Interpolatory rational model order reduction of parametric problems lacking uniform inf-sup stability”, ENUMATH 2019, Egmond aan Zee (NL).
 2021 [DP, F. Nobile, and F. Bonizzoni, “Non-intrusive model reduction of parametric frequency response problems via minimal rational interpolation”, ICOSAHOM 2020/2021 (virtual), Vienna (AT).
 [DP and F. Nobile, “Non-intrusive model reduction of parametric frequency-response problems – with applications to UQ”, SIMAI 2020+2021, Parma (IT).
 2022 [DP and F. Nobile, “Non-intrusive surrogate modeling of parametric frequency response problems – With applications in forward UQ”, SIAM UQ22 (virtual), Atlanta (Georgia, US).
 DP and F. Nobile, “Inexpensive surrogate modeling of frequency response problems by greedy minimal rational interpolation”, GAMM Annual Meeting 2022, Aachen (DE).
 [DP and F. Nobile, “Non-intrusive surrogate modeling of frequency response surfaces via locally adaptive sparse grids”, GIMC SIMAI Young 2022, Pavia (IT).
 2023 [DP, M. Nonino, and I. Perugia, “Geometry-based approximation of waves propagating through complex domains”, 17th Austrian Numerical Analysis Day, Vienna (AT).
 DP, F. Nobile, and A. Borghi, “Non-intrusive surrogate modeling of parametric frequency-response problems”, Math2Product 2023, Taormina (IT).
 DP, “Rational approximation with minimal sampling for Helmholtz-like problems”, ILAS 2023, Madrid (ES).
 [DP, F. Nobile, and A. Borghi, “Data-driven adaptive approximation of parametric dynamical systems with pole bifurcations”, ENUMATH 2023, Lisbon (PT).
 2024 [DP, F. Nobile, and A. Borghi, “Adaptive collocation-based approximation of parametric non-linear eigenproblems”, SIAM UQ24, Trieste (IT).

- [DP and A. Borghi, “Adaptive collocation-based approximation of parametric nonlinear eigen-problems”, ECM 2024, Seville (ES).
- 2025 [DP, A. Borghi, A. Kleefeld, and L. Pieronek, “Approximation of parametric eigenvalues – with applications to interior transmission eigenvalue problems”, NACONF 25, Glasgow (UK).

Posters

- 2018 [F. Bonizzoni, I. Perugia, F. Nobile, and DP, “An efficient algorithm for Padé-type approximation of the frequency response for the Helmholtz problem”, MoRePaS IV, Nantes (FR).
- [F. Bonizzoni, I. Perugia, F. Nobile, and DP, “An efficient algorithm for Padé-type approximation of the frequency response for the Helmholtz problem”, Swiss Numerics Day 2018, Zurich (CH).
- 2020 [DP and F. Nobile, “Frequency-domain non-intrusive greedy Model Order Reduction based on minimal rational approximation”, SCEE 2020, Eindhoven (NL).
- [DP and F. Nobile, “Frequency-domain non-intrusive greedy Model Order Reduction based on minimal rational approximation”, MORSS 2020 (virtual), Lausanne (CH).
- 2022 [DP and F. Nobile, “Non-intrusive adaptive surrogate modeling of parametric frequency-response problems”, MORE 2022, Berlin (DE).

Other talks and seminars

- 2018 [DP, F. Nobile, F. Bonizzoni, and I. Perugia, “Fast Least-Squares Padé approximation of self-adjoint problems with meromorphic structure”, seminar, MATHICSE retreat, Sainte-Croix (CH).
- [DP, F. Nobile, F. Bonizzoni, and I. Perugia, “Fast Least-Squares Padé approximation of self-adjoint problems with meromorphic structure”, workshop talk, DRWA, Alba di Canazei (IT).
- 2019 [DP and F. Nobile, “Polynomial approximation of resonance manifolds”, short seminar, MATH-ICSE retreat, Champéry (CH).
- 2020 [DP, “Padé approximation: a quick overview”, seminar (virtual), CSQI talks, Lausanne (CH).
- [DP, “From Padé approximation to rational interpolation”, seminar (virtual), CSQI talks, Lausanne (CH).
- [DP, “Minimal rational approximation”, seminar (virtual), CSQI talks, Lausanne (CH).
- [DP, “Minimal rational approximation: a model reduction tool for parametrized PDEs with resonances”, seminar (virtual), PDE afternoons, Vienna (AT).
- 2021 [DP, “Matching-based pMOR for dynamical systems”, seminar (virtual), CSQI talks, Lausanne (CH).
- 2022 [DP, “Surrogate modeling of parametric frequency response problems via locally adaptive sparse grids”, workshop talk, “Approximation of high-dimensional parametric PDEs in forward UQ” ESI workshop, Vienna (AT).
- [DP, “Can reliable surrogate models for frequency-domain problems be both non-intrusive and cheap to build?”, workshop talk, “UQ in kinetic and transport equations and in high-frequency wave propagation” ESI workshop, Vienna (AT).
- 2023 [DP, “Efficient and adaptive rational approximation for parametric dynamical systems”, seminar, CSC seminar, Magdeburg (DE).
- 2024 [DP, “Adaptive data-driven surrogate modeling of parametric nonlinear eigenproblems”, seminar, NA seminar, KTH, Stockholm (SE).

- 2025 [DP, “Numerical methods for surrogate modeling”, short seminar, SFG division meeting, KTH, Stockholm (SE).
- DP, “Numerical methods for surrogate modeling”, open-science seminar, Math department staff meeting, SU, Stockholm (SE).
- [DP, “Chasing resonances with adaptive rational approximation”, seminar, ZHACM seminar, ETH Zurich, Zurich (CH).

Attended events

- 2018 [“Numerical Analysis of Complex PDE Models in the Sciences” ESI workshop, Vienna (AT).
- 2021 [Swiss Numerics Day 2021, Lausanne (CH).
- 2022 [“Adaptivity, High Dimensionality and Randomness” ESI workshop, Vienna (AT).
- Austrian Numerical Analysis Day 2022, Linz (AT).
- [MCQMC 2022, Linz (AT).
- 2023 [“Canonical scattering problems” INI workshop, Cambridge (GB).
- [2nd SFB International Workshop 2023, Vienna (AT).
- 2024 [“Celebrating Zhaojun Bai’s honorary doctorate” workshop, Stockholm (SE).
- [5th Workshop on Scientific Computing in Sweden, Stockholm (SE).
- 2025 [“Advances in Stochastic Processes and Numerical Analysis - A Marcus Wallenberg Symposium”, Stockholm (SE).
- “Stabilized and Cut Finite Element Methods – Celebrating Peter Hansbo” IML workshop, Stockholm (SE).
- [“Interfaces and Unfitted Discretization Methods” IML workshop, Stockholm (SE).
- 2026 [3rd Digital Futures Young Scientist Conference, Stockholm (SE).
- [17th SeRC Annual Meeting, Lidingö (SE).

Teaching experience

- 2017–2021 [As teaching assistant at EPFL (preparation of course and exercise material, preparation and grading of assignments and exams):
 - 2017 (Analyse avancée I, BSc in Mathematics.
 - 2018 (Analyse numérique, BSc in Mechanical Engineering.
 - 2018 (Analyse fonctionnelle, BSc in Mathematics.
 - 2019 (Introduction to partial differential equations, BSc in Mathematics.
 - 2021 (Numerical analysis and computational mathematics, MSc in Computational Sciences and Engineering.
- 2019–2021 (Parallel and high-performance computing, MSc in Computational Sciences and Engineering.
- 2022 [Invited lecturer for: Numerical methods for random PDEs and uncertainty, PhD course, EPFL.
- 2022–2023 [As university assistant at U Vienna (charged with the organization of the whole course):
 - 2022–2023 (Exercises of Analysis and Linear Algebra I, BSc in Mathematics.
 - 2023 (Topics in Finite Elements, MSc in Mathematics.
 - 2023 (Exercises of Applied Mathematics, BSc Teacher Training Program in Mathematics.
 - 2023 (Exercises of Numerical Mathematics, BSc in Mathematics.
- 2024–2025 [As postdoc at KTH (co-responsible of the whole course):
 - 2025 (Numerical Methods for Differential Equations II, MSc in Mathematics.

Supervision and service on grading committees

- 2019 [Supervision of BSc thesis: “Approximation numérique du spectre des opérateurs elliptiques d’ordre deux” by T. Chanay, EPFL.
- 2022 [Supervision of MSc project: “Minimal rational approximation for time-harmonic Maxwell’s equations” by F. Matti, EPFL.
- 2025 [Supervision of MSc thesis: “Kolmogorov Arnold Networks for differential applications within atmospheric modelling” by S. Vogel Hauger, KTH.
 - [Assessment of MSc thesis: “Utilizing multibeam bathymetry data in 3D-mesh generation for finite element simulations of glacier-fjord circulation” by V. Bäcklin, SU.
- 2026 [Half-time assessment of PhD project: “Beyond exchangeability in conformal prediction” by J. Hallberg Szabadváry, JU.

Teaching education

- 2023 [“When to Teach What? - Sequencing Course Content and Assignments”, Center for Teaching and Learning, University of Vienna.
 - “(Unconscious) Bias at the University”, Center for Training and Development, University of Vienna.
 - [“Introduction to Active Learning”, Center for Teaching and Learning, University of Vienna.

Other service

- 2020 [Organizer of the Model Order Reduction Summer School 2020 (virtual event).
 - [Referee for scientific journals: “Advances in Computational Mathematics”.
- 2022 [Referee for scientific journals: “Journal of Computational Physics”.
- 2023 [Chairperson for “Data-driven reduced order modelling and surrogates with applications in complex multi-physics systems” mini-symposium at M2P 2023.
 - Organizer of the “Reducing the irreducible: model reduction for transport-dominated problems” mini-symposium at ENUMATH 2023.
 - [Referee for scientific journals: “Advances in Computational Mathematics”, “Computational Methods in Applied Mathematics”, “Journal of Computational Physics”, “Inverse Problems”, and “SIAM Journal of Scientific Computing”.
- 2024 [Organizer of the “Methods for stochastic/parametric frequency-domain and spectral problems” mini-symposium at SIAM UQ24.
 - [Referee for scientific journals: “Advances in Computational Mathematics”, “Applied Mathematics Letters”, “Computational Methods in Applied Mathematics”, “IEEE Transactions on Microwave Theory and Techniques”, and “Mathematics and Computers in Simulation”.
- 2025 [Organizer of the “Career development activities for PhDs and postdocs in Math” series at KTH.
 - [Referee for scientific journals: “Applied Mathematics and Computation”, “IMA Journal of Numerical Analysis”, “Journal of Computational Physics”, “Journal of Scientific Computing”, and “23rd European Control Conference”.
- 2026 [Organizer of the “Recent advances in computational uncertainty quantification for complex PDEs” mini-symposium at CMAM 2026.
 - Organizer of the “Advances in linear algebra and surrogate modeling for scattering problems” mini-symposium at Wave-Ned 2026.

- 2024–now [Referee for scientific journals: “Applied Mathematics and Computation”, “IMA Journal of Numerical Analysis”, and “SIAM Journal of Scientific Computing”.
- [Editorial board member of “Examples and Counterexamples”.
- [Chair of the Numerical Analysis seminar series at KTH&SU.

Computer skills

- Advanced [MATLAB, C/C++, OpenMP, MPI, Python, L^AT_EX.
- Intermediate [CUDA, C#, FreeFem++, HTML, CSS.
- Basic [R, OpenFOAM, Fluent, Fortran, Java.

Languages

Italian:	Native	English:	Fluent
Swedish:	Intermediate	Spanish:	Intermediate
French:	Intermediate	German:	Basic

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